MIL-C-81812A
1 October 1971
Superseding
MIL-C-81812(AS)
10 February 1971

MILITARY SPECIFICATION

CLOTH, KNITTED, AROMATIC POLYAMIDE, HIGH TEMPERATURE RESISTANT

* This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the procurement of a high temperature resistant aromatic polyamide staple fiber knitted cloth used in aeronautical clothing.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D. C. 20580,)

3. REQUIREMENTS

3.1 Standard sample - The cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 Materials - The cloth shall be knit from a high strength, non-melting, aromatic polyamide staple yarn. The yarn shall not char at a temperature less than 675°F (357°C) when tested as specified in 4.3.

* 3.3 Color - Unless otherwise specified, the color of the cloth shall be as specified by the procuring activity. The color shall be obtained by the use of melt spun solution dyed fibers.

3.3.1 Matching - The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

* 3.3.2 Colorfastness - The dyed and finished cloth shall show fastness to light and to laundering equal to or better than the standard sample when tested as specified in 4.3. When determining lightfastness, the sample shall be faded to the break point of the standard reference sample and the supplier's submission shall be compared with the standard at this point.

3.4 Physical requirements - The finished cloth shall conform to the requirements specified in Table I.

3.5 Knit - The cloth shall be 1 by 1 rib and knit on any type of knitting machine which will produce a cloth having desired characteristics and meet the requirements specified herein.

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TABLE I

* PHYSICAL PROPERTIES

CHARACTERISTICS	REQUIREMENTS
Staple denier, nominal	2
Staple length, inch	1.5 - 2.0
Weight, oz. per square yard, min.	12.0
Texture per inch each side, min.	
Wale	16
Course	26
Acidity, pH	5.0 - 8.0

* 3.6 Width - The width of the finished cloth shall be as specified which shall be the minimum acceptable width inclusive of selvage.

3.7 Length and put-up - Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on rolls as specified in PPP-P-1133.

3.8 Fiber identification - Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship - The finished cloth shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance - Where certificates of compliance are submitted in accordance with 4.3, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the cloth to verify the validity of the certification.

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4.2 Inspection - Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Examination of the end item - Defects found during this examination shall be classified in accordance with 4.2.1.1 through 4.2.3.4.

* 4.2.1.1 Yard-by-yard examination - The required yardage of each roll shall be examined on the face side and visual defects classified as listed in Table II. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (length-wise) yard or fraction in which it occurs. The sample unit shall be one linear yard. The Acceptable Quality Levels shall be 2.5 major defects and 6.5 total defects per 100 units (yards). The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with Table III. An approximately equal number of yards shall be examined from each roll in the sample.

TABLE II

CLASSIFICATION OF DEFECTS

DEFECTS	MAJOR	MINOR
Hole, cut, tear, run or dropped stitch	X	
Tuck stitch, pulled stitch or needle streak		X
Mend (any)	X	
Uneven, ridge or cockled cloth		X
Thin or uneven area	X	
Abrasion mark resulting in a weak area	X	
Slub, slug or gout, more than 3 times the normal yarn thickness		X
Coarse or fine yarn 1/		X
Tight twist yarn		X
Foreign matter 1/		X
Permanent crease in cloth	X	
Knots, 2 or more, extending above surface of cloth		X
Kinks or untrimmed ends		X
Stretch (stitch tension) knitting tension too tight or too loose	X	
Shade bar 1/	X	
Spot or stain		X

1/ Clearly visible at normal inspection distance (approximately three feet).

4.2.1.2 Overall examination - Each defect shall be counted not more than once in each roll examined. The sample units shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be as shown in Table III.

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TABLE III

SAMPLE SIZE

LOT SIZE (YARDS)	SAMPLE SIZE (ROLLS)	MAXIMUM NUMBER OF DEFECTS ACCEPTABLE IN SAMPLE
Up to 1200	3	0
1201 up to and including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

DEFECTS

Width less than specified.
Overall uncleanness.
Uneven knitting throughout the roll.

4.2.1.3 Examination for length -

4.2.1.3.1 Individual rolls - The rolls shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table III.

4.2.1.3.2 Total yardage in sample - The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.1.3.3 Examination for compliance with Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

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4.2.1.3.4 Examination for shade - During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.2.2 Examination of preparation for delivery requirements - Examination shall be made in accordance with the provision of PPP-P-1133 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.3 Testing of the end item - The methods of testing specified in FED-STD-191 wherever applicable shall be followed and as listed in Table IV shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 2 continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual value utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with Table V.

TABLE IV

TEST METHODS

CHARACTERISTICS	REQUIREMENT REFERENCE	TEST METHOD
Identification	3.1	1/
Char Point	3.1	1/
Staple denier, nominal	Table 1	visual 1/
Staple length, inch	Table 1	visual 1/
Weight, oz. per square yard, min.	Table 1	5041
Texture per inch, min.	Table 1	
Wale		5070
Course		5070
Acidity, pH	Table 1	2811
Ply	3.2	4020 or 4021
Colorfastness:		
Light	3.3.2	5660 2/
Laundering	3.3.2	5610
Type of Knit	3.5	Visual

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1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Except that the supplier's submission shall be compared with the standard sample after 6 hours and evaluation made at this point.

TABLE V

SAMPLING FOR TESTING

LOT SIZE (UNITS)	SAMPLE SIZE
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

* 5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging - Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C - The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use - The cloth covered by this specification is intended for use in cuffs, anklets, and waistband for flight clothing.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Width of cloth required (see 3.6).

(c) Minimum length if other than specified (see 3.7).

(d) Selection of applicable levels of packaging and packing (see 5.1).

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(e) Color of cloth required (see 3.3).

* 6.3 The knitted cloth requirements were based on a construction using two-ply yarn each equal to 26s worsted count.

6.4 Standard sample - For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Air Force - 11

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Preparing Activity:
Navy - AS

Project No. 8305-0178

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-81812A

CLOTH, KNITTED, AROMATIC POLYAMIDE, HIGH TEMPERATURE RESISTANT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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